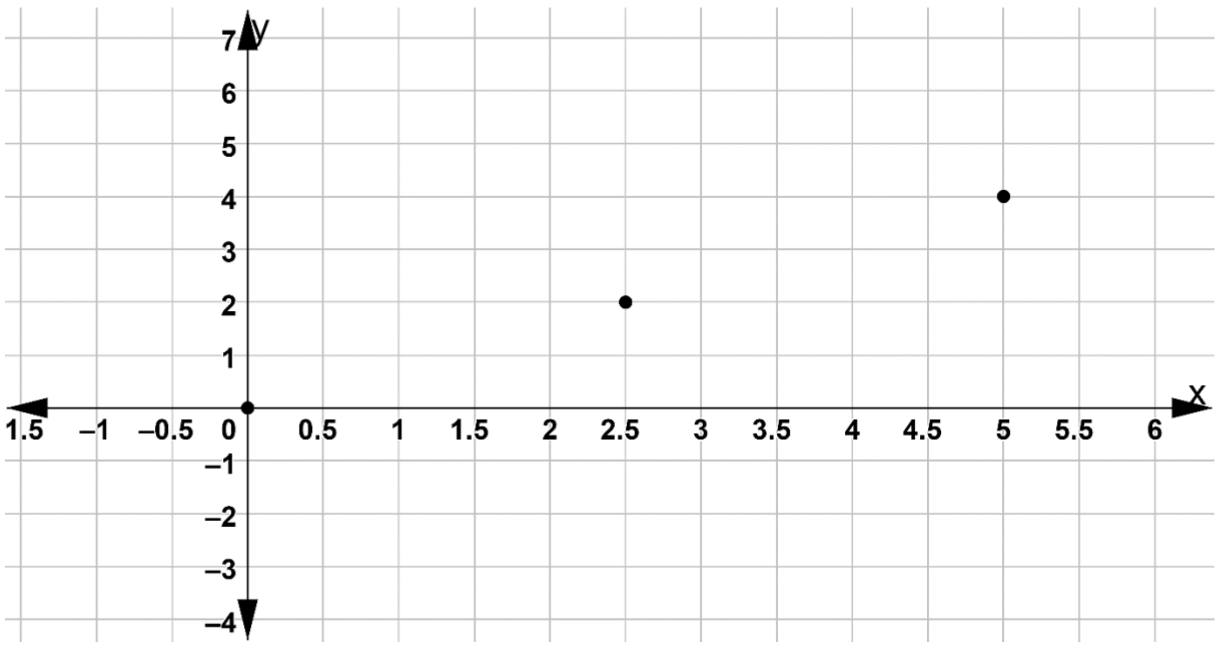
Grade 7 Name____________________
Unit 11 Assessment Date_________________
Calculator Period_________

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result.

1. A proportional relationship is shown on the coordinate grid.



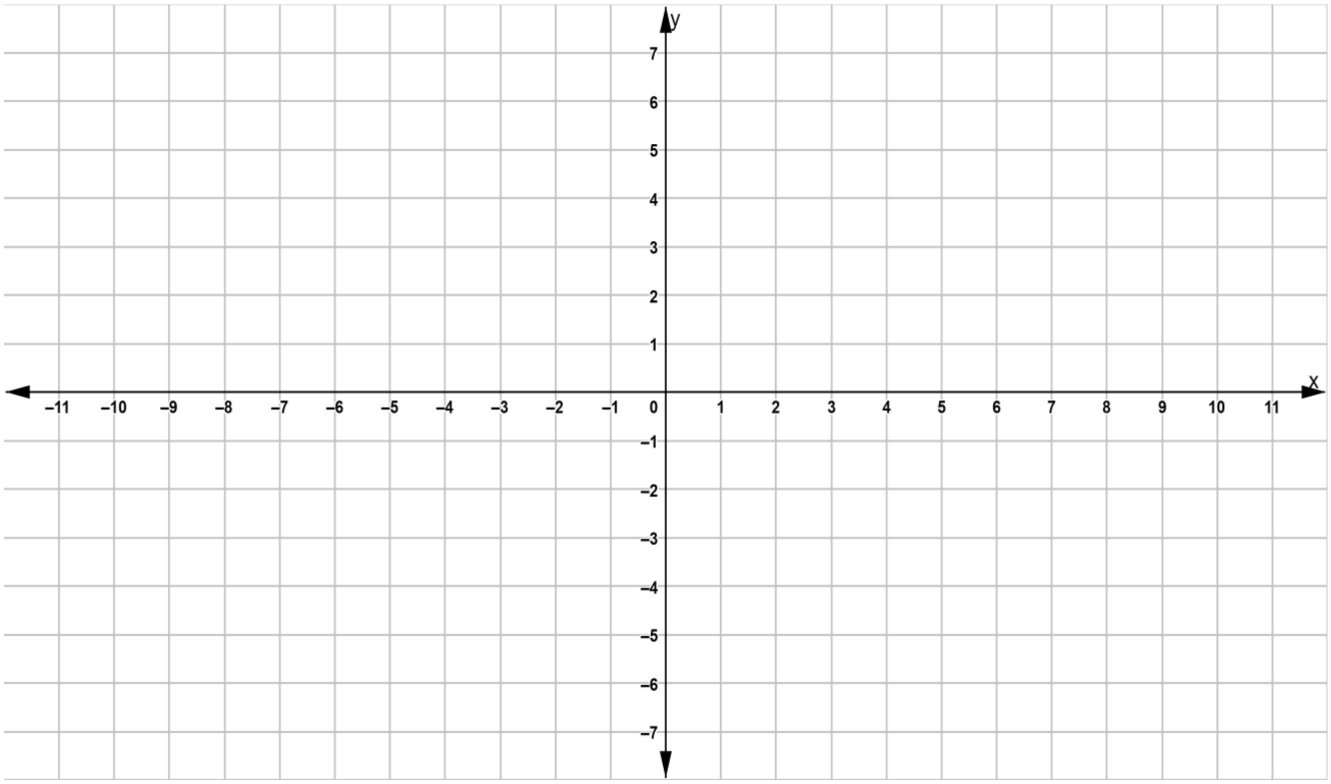
Determine the constant of proportionality.

work	
Constant of Proportionality	

2. The equation of a proportional relationship is shown.

$$y = \frac{3}{5}x$$

Graph the proportional relationship on the coordinate grid.



3. The principal at Carr Elementary and Middle School bought candy bars to be given to teachers for Teacher Appreciation Week. Her first order was for 36 candy bars and she paid $\$27$. The second order was for 24 candy bars and she paid $\$18$. The relationship between candy bars and the amount paid is proportional. Write an equation to represent the relationship, where y is the number of candy bars and x is the cost, in dollars.

work	
Equation	

4. Saffron is the world's most expensive spice by weight. Saffron is used in recipes such as Spanish Paella or Arroz con Pollo. It requires tweezers to harvest the part of the saffron crocus that is used to make the spice. For this reason, harvesting saffron takes a long time. The table shows the proportional relationship between time, in hours, it takes to harvest and the amount of saffron harvested, in ounces.

Time (hours)	420	840	1,008	1,260
Saffron (ounces)	35	70	84	105

a. Determine the constant of proportionality.

work	
Constant of Proportionality	

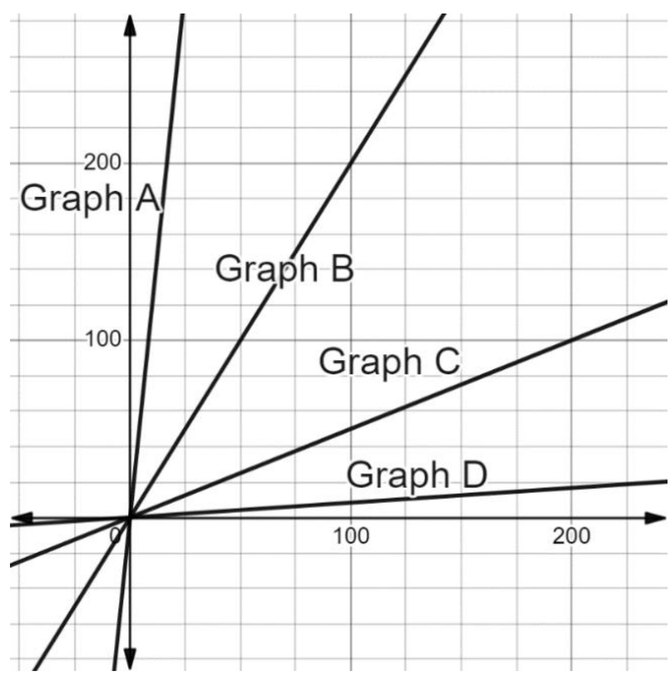
b. Write an equation to represent the relationship between the amount of saffron, y, in ounces, and the time it takes to harvest, x, in hours.

Equation	

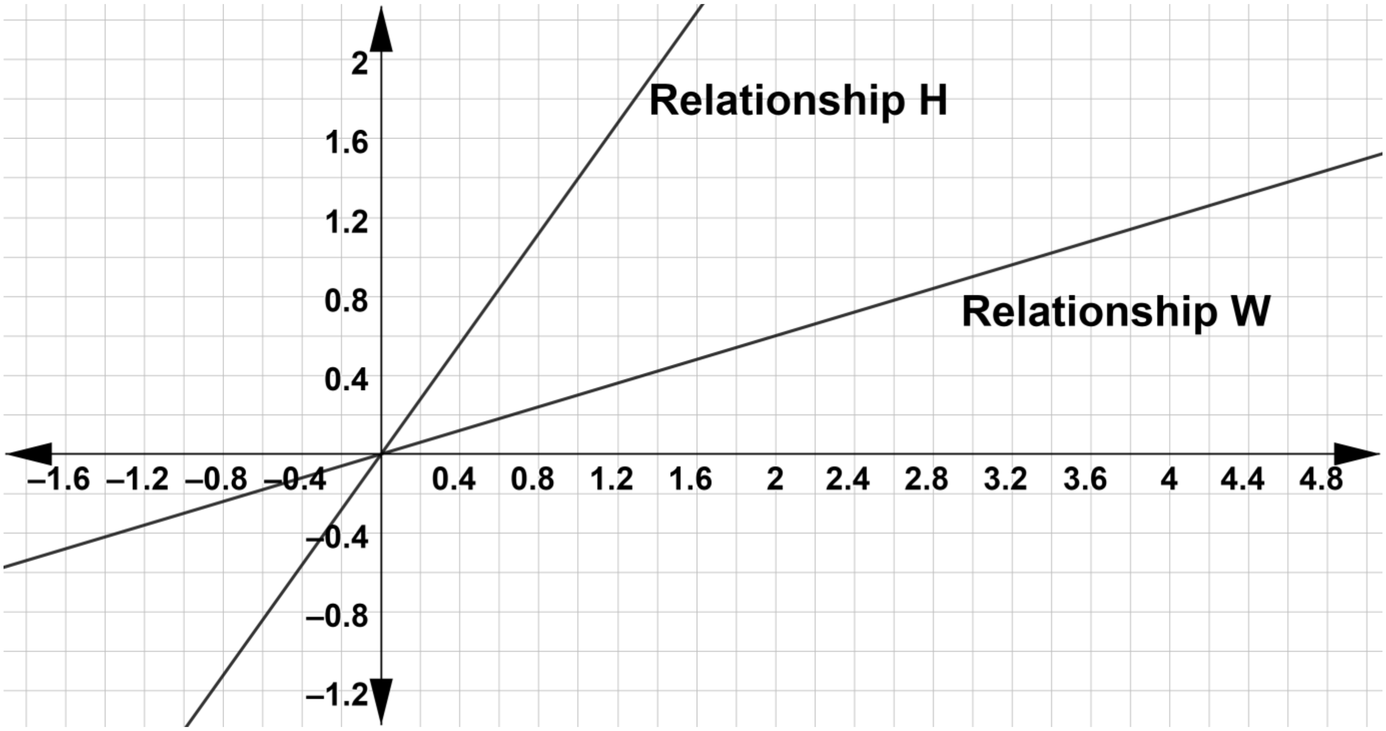
c. Four proportional relationships are shown on the coordinate grid.

Which graph shows the proportional relationship between the time, x, in hours, it takes to harvest and the amount of saffron harvested, y, in ounces.

Graph	



5. Two proportional relationships are shown on the coordinate grid.



a. Complete the table of values for relationship H.

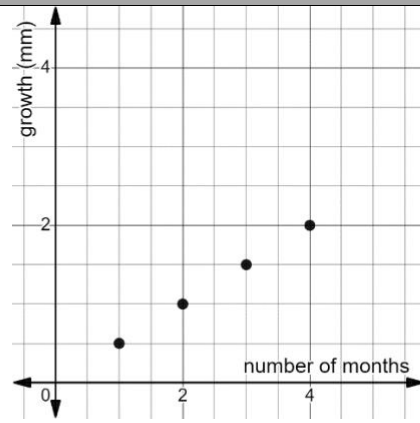
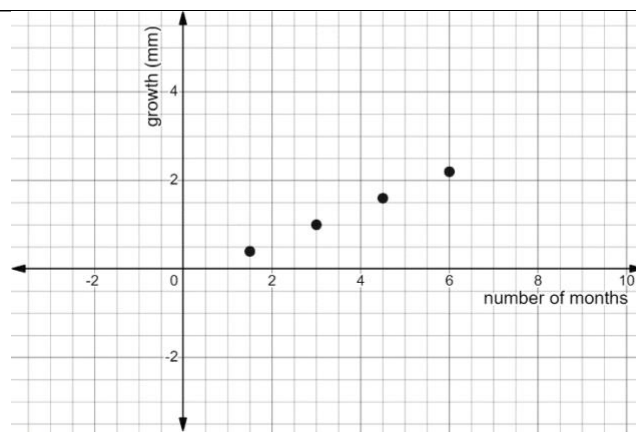
x	0	1	2	
y	0			7

b. Write an equation to represent relationship W.

work

Equation	

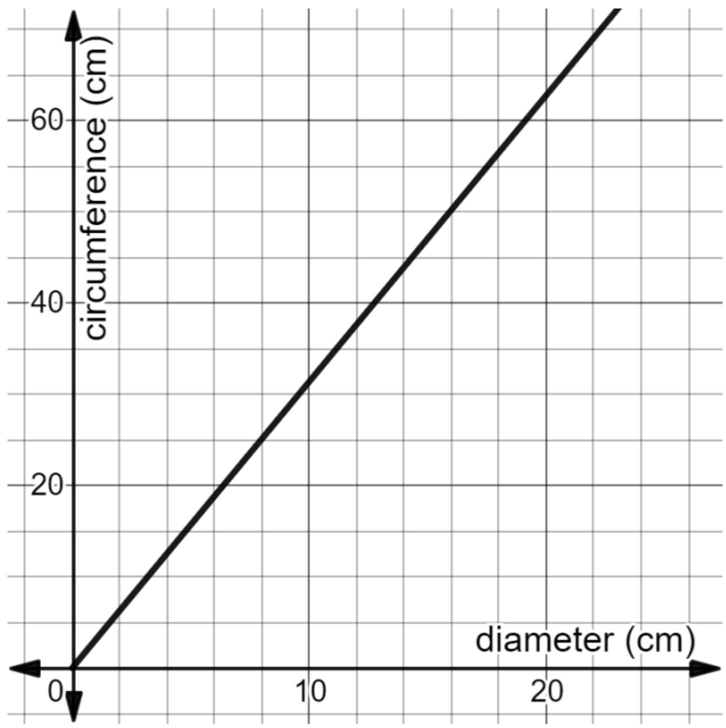
6. The Florida perforate reindeer lichen is found exclusively in Florida. A botanist was conducting research on this endangered lichen and monitored the growth, in millimeters (mm), of the lichen in different locations, as shown.

<table>
<tr><th>Location</th><th>Growth</th></tr>
<tr><td>Santa Rosa Island</td><td>
</td></tr>
<tr><td>Jonathan Dickinson State Park</td><td><table><tr><th>Number of Weeks</th><td>3</td><td>7</td><td>11</td><td>20</td></tr><tr><th>Growth (mm)</th><td>0.3</td><td>0.6</td><td>0.9</td><td>1.2</td></tr></table></td></tr>
<tr><td>Lake Wales Ridge National Wildlife Refuge</td><td>On May 3rd, growth was 0.11 mm. On May 17th, growth was 0.33 mm. On May 24th, growth was 0.44 mm.</td></tr>
<tr><td>Tiger Creek Preserve</td><td><table><tr><th>Number of Months</th><td>3</td><td>7</td><td>11</td><td>20</td></tr><tr><th>Growth (mm)</th><td>0.39</td><td>0.91</td><td>1.43</td><td>2.6</td></tr></table></td></tr>
<tr><td>Kissimmee Prairie Preserve State Park</td><td>
</td></tr>
</table>

Select all of the locations where the growth of the lichen was proportional.

- ☐ Santa Rosa Island
- ☐ Tiger Creek Preserve
- ☐ Jonathan Dickinson State Park
- ☐ Kissimmee Prairie Preserve State Park
- ☐ Lake Wales Ridge National Wildlife Refuge

7. Christiano measured the circumference and diameter of several different circular objects using centimeters (cm). He then used the information to generate the graph shown.



Complete the statements.

The graph shows that the relationship between the circumference and the diameter is

- ○ proportional.
- ○ not proportional.

The constant of proportionality

- ○ is 2.
- ○ is 3.
- ○ is π.

The equation that represents the proportional relationship is

- ○ $y = 2x$.
- ○ $y = \frac{3}{x}$.
- ○ $y = \pi x$.
- ○ $y = \pi x^2$.